

Nizar Esmat Fawzy

Software Engineer

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Work Experience

Opream

Software Engineer

May 2025 – Present

Cairo, Egypt

- Own and deliver production features end-to-end across the stack — React, NestJS, PostgreSQL.
- **Mostaqbel Watan Platform** — private social platform for org members with Jitsi-integrated video meetings, Socket.io private/group chat, and a Facebook Graph API integration to auto-ingest posts from a designated group.
- **Onway** — ride-share platform connecting riders and drivers, integrated with the Fawry payment gateway.
- **Bingo** — library management system.
- **Al-Azhar** — unified platform managing online and offline student lessons.
- **Event Ticketing Platform** — implemented a scheduled job to auto-release stale "pending" ticket reservations back to "new" every 10 minutes, preventing inventory lock-up.

Information Technology Institute (ITI)

Full Stack Developer Intern (Python)

Nov 2024 – Mar 2025

Smart Village, Egypt

- Designed a normalized PostgreSQL schema supporting 10k+ users with data integrity, indexing, and query optimization.
- Designed and delivered RESTful APIs with ETL-style data workflows, contributing to architecture decisions and cross-team integration across the full development lifecycle.

iSchool & WE School

Software & Web Development Instructor (Part-time)

Apr 2024 – Nov 2024

Cairo, Egypt

- Taught programming fundamentals, mobile development (Scratch), and software problem-solving to students.
- Taught web development (HTML, CSS, JavaScript) and software problem-solving to students.

Projects

Saraha App | Node.js, Express.js, MongoDB, JWT, bcryptjs, CryptoJS

GitHub ↗

- Architected a secure anonymous messaging REST API with JWT email verification, bcrypt hashing, and CryptoJS AES encryption for sensitive data.
- Built a service-layer architecture with Joi validation and a global asyncHandler error handler; implemented soft-delete account freezing with automatic restoration instead of hard deletes.

Chat Application | React, Node.js, Express.js, Socket.io, MongoDB, Cloudinary

GitHub ↗

Postman ↗

- Engineered a full-stack real-time chat app (React/Zustand, Node/Express/Socket.io) with private messaging, group channels, live presence, and live message edit/delete sync.
- Secured the API with HTTP-only cookie JWT auth and Arcjet rate limiting against spam/abuse; integrated Cloudinary for image/voice storage with full CRUD.

ITI Social Media Platform | PostgreSQL, REST API, React

GitHub ↗

- Co-led backend architecture for a full-stack social platform: designed normalized PostgreSQL schema, built RESTful APIs, and integrated with a React frontend.
- Drove the full SDLC: requirements analysis, database modeling, backend development, and cross-team integration.

Education

Nahda University

Bachelor of Computer Science

Sep 2019 – Jul 2023

Beni-Suef, Egypt

- Graduation Project: Smart Learning Management System with a Machine Learning model.

Technical Skills

Languages: JavaScript, Python, C++, SQL
Frameworks: React, Node.js, NestJS, Express.js, Django (familiar), Angular (familiar)
Databases: PostgreSQL, MongoDB, Database Design, Query Optimization, Data Modeling, ETL
Software Engineering: SDLC, REST APIs, OOP, System Architecture, Real-Time Apps, Authentication
Security: CORS, XSS, CSRF mitigation, Input Validation
Tools: Git, GitHub, Docker, Linux, Bash, CI/CD, SAS (basic familiarity)

Activities & Languages

Arabic (Native), English (Fluent) | ECPC/ACPC/ICPC Operations Volunteer, 2022–2024 | ECPC Contestant 2021 & 2022, 2nd place

Courses

Python Professional Track, Information Technology Institute (ITI) | SQL and PostgreSQL – The Complete Developer’s Guide (Udemy) | Node.js – The Complete Guide (Udemy) | Full Stack Web Development (MERN), Route Academy